DEEP LEARNING LABORATORY

EXPERIMENT 8

TWO-LAYER NEURAL NETWORK WITH BACKPROPAGATION

Aim

To implement a two-layer neural network using forward propagation and backpropagation, train it on the XOR dataset, and observe the reduction in binary cross-entropy loss.

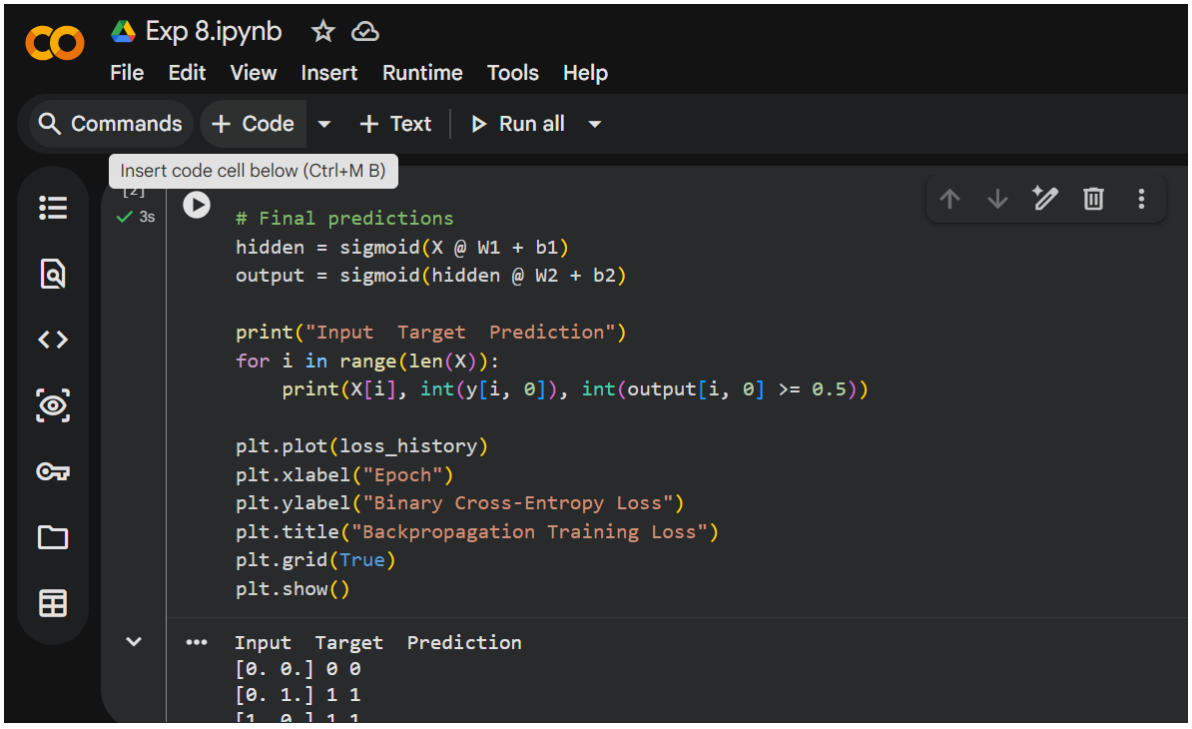
Output

The network calculates predictions for the four XOR input combinations. The training-loss graph shows the binary cross-entropy loss decreasing during training, indicating that the network is learning the XOR pattern.

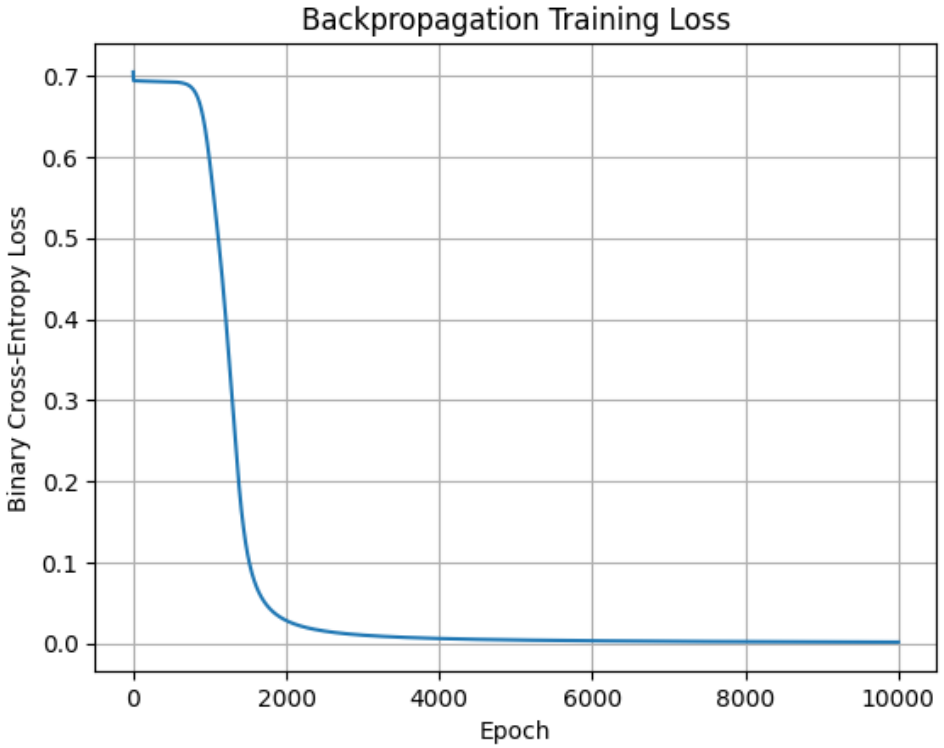
Observed output

The screenshot shows the final predictions being printed for the XOR inputs. The visible rows include [0, 0] → 0, [0, 1] → 1, and [1, 0] → 1; the final [1, 1] row is cut off in the supplied screenshot, so it is not inferred here.

Colab output screenshot



Training-loss graph


Backpropagation Training Loss
Binary Cross-Entropy Loss
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
2000
4000
6000
8000
10000
Epoch